👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 · STANDARD 6.AT.11

Apply Two-Variable Relationships to Solve Problems

Lesson 9-4 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write an equation to represent the relationship between two variable quantities and use it to solve a real problem.

Language Objective: I can justify a recommendation using the words equation, solution, and at least.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

solution · substitute · at least · predict · justify · Equation · Inverse operation · Work backward

1. A value of the variable that makes an equation true is a _____.

2. To replace a variable with a number so you can compute is to _____.

3 That amount or more — the smallest amount that still works — is _____ _____ .

4 To use what you already know to say what a future amount will be is to _____ .

5 To explain WHY your answer makes sense using the mathematics is to _____ it.

6 A math sentence with an equals sign is an _____ .

7 Division is the _____ of multiplication.

8 Starting from the output and undoing steps to find the input is called _____ .

Write About the Math

From which bakery would you recommend Yelina purchase cupcakes? Explain?
¿De qué panadería le recomendarías a Yelina comprar los panquecitos? Explica?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **solution** — *solución*
- ☐ **substitute** — *sustituir*
- ☐ **at least** — *al menos*
- ☐ **predict** — *predecir*
- ☐ **justify** — *justificar*

Model: *Since h is being multiplied by 2.5 in the equation, dividing by 2.5 is the operation that undoes the multiplication and isolates h.* — A strong answer names division as the inverse of multiplication and checks the solution by substituting it back in. A weak answer just says 'because that's what you do' without the inverse-operation reasoning.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Write an ___ for each bakery, substitute Yelina's $___ budget into each one, ___ the two results, and recommend the bakery that gives more ___ for the money.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** solution
2. **Notes 2:** substitute
3. **Notes 3:** at least
4. **Notes 4:** predict
5. **Notes 5:** justify
6. **Notes 6:** Equation
7. **Notes 7:** Inverse operation

8. **Notes 8:** Work backward
9. **Watch for this mistake:** *Adding the rate instead of multiplying — writing 2.5 + 8 = 10.5 cars for an 8-hour shift instead of 2.5 × 8 = 20. The rate applies to EVERY hour, so the total is always rate × amount.*
10. **Try It 1:** A. No — cars come in whole numbers, so round up to the next whole car — *You cannot wash seven-tenths of a car, so the model's decimal answer must be interpreted: round up so the goal is still met.*
11. **Try It 2:** A. Either can work: $12 needs at least 42 cars washed, while $14.30 needs only about 35 — so the choice depends on how many customers they expect — *Both plans reach $500 — 42 × 12 = $504 and 35 × 14.30 = $500.50 — so the real question is whether they can attract 42 cars at the lower price, or whether the higher price might turn customers away.*